

“Symptom-Based Disease Prediction Framework Integrated with Django and Deep Learning Models”

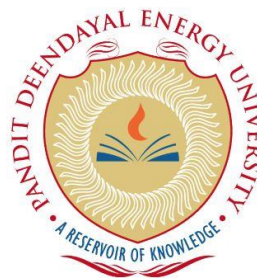
Comprehensive Project Report

*Submitted in Partial Fulfillment of the
Requirements for the Degree of*

BACHELOR OF TECHNOLOGY IN INFORMATION AND COMMUNICATION TECHNOLOGY ENGINEERING

By
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May 202

Certificate of Originality of Work

I hereby declare that the B.Tech. Project entitled “Symptom-Based Disease Prediction Framework Integrated with Django and Deep Learning Models” submitted by me for the partial fulfillment of the degree of Bachelor of Technology to the Dept. of Information and Communication Technology Engineering at the School of Technology, Pandit Deendayal Energy University, Gandhinagar, is the original record of the project work carried out by me under the supervision of Dr. Soumyashree Panda.

I also declare that this written submission adheres to University guidelines for its originality, and proper citations and references have been included wherever required.

I also declare that I have maintained high academic honesty and integrity and have not falsified any data in my submission.

I also understand that violation of any guidelines in this regard will attract disciplinary action by the institute.

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Certificate from the Project Supervisor/Head

This is to certify that the Major/Comprehensive Project Report entitled “**Symptom-Based Disease Prediction Framework Integrated with Django and Deep Learning Models**” submitted by **Ms. Vanshita Patel** , Roll No. **20BIT157D** towards the partial fulfilment of the requirements for the award of degree in Bachelor of Technology in the field of **Information and Communication Technology** Engineering from the **School of Technology**, Pandit Deendayal Energy University, Gandhinagar is the record of work carried out by him/her under my/our supervision and guidance. The work submitted by the student has in my/our opinion reached a level required for being accepted for examination. The results embodied in this major project work to the best of our knowledge have not been submitted to any other University or Institution for the award of any degree or diploma.

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Vanshita Patel

Abstract

In healthcare, spotting diseases early is crucial. This project uses machine learning to predict diseases based on symptoms. We used CSV files, which contain a dataset with lots of symptoms and diseases. Our fancy model, the Convolutional Neural Network (CNN) Sequential Model with SoftMax activation function, looks at these symptoms to find patterns and identify diseases early.

We're trying to predict all sorts of diseases like Fungal infection, Allergy, Diabetes, etc. covering a wide range of health issues. By studying different symptoms like Itching, Skin rash, Nodal skin Eruptions, etc. our model learns how to tell disease possibility from it.

After lots of training and testing, our model can predict diseases with an impressive accuracy of 97.66%. But it's not just about finding diseases – we want to give doctors a tool that can also help them understand how diseases progress and what might happen next. To make this process accessible and user-friendly, we have designed a user interface (UI) that simplifies interaction with the model. Through this interface, users can input their symptoms, and the model provides predictions about potential diseases. Moreover, users have the option to connect with healthcare professionals for further guidance and treatment.

This project breaks new ground by utilizing symptom-based data for disease prediction, aiming to detect illnesses earlier and expedite treatment. Through the integration of advanced technology, comprehensive symptom data, and a user-friendly interface (UI), our goal is to enhance healthcare accessibility and outcomes for all.

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Chapter 1

Introduction

1.1 Prologue

Our study takes a thorough approach, exploring the depths of past medical data, patient signs and symptoms, and pertinent health measures to identify trends and revelations that are essential for precise illness prediction. By utilizing Django, a potent high-level Python web framework, we are able to create a flexible and reliable web application interface. Our platform's core is this interface, which efficiently manages data and makes user interaction simple and accurate.

At the heart of our endeavor lies the deployment of sophisticated machine learning algorithms. These algorithms serve as the backbone of our predictive model, tirelessly sifting through vast troves of data to extract meaningful insights and relationships. By leveraging advanced techniques and methodologies, our aim is to elevate the accuracy and reliability of disease prediction, thereby empowering both healthcare providers and patients with valuable foresight and actionable intelligence.

The platform we offer represents a beacon of convenience and accessibility in the realm of healthcare. It extends a virtual lifeline to users, granting them the freedom to access medical advice and assistance from the comfort of their own homes. Through intuitive user interfaces and interactive features, individuals can seamlessly input their symptoms, receive personalized predictions tailored to their unique health profile, and even connect with healthcare professionals for further guidance and support.

Beyond mere convenience, our project aspires to bridge the gap between technology and healthcare, catalyzing a paradigm shift towards a more patient-centric approach. By empowering individuals to take charge of their health journey and seek timely intervention, we envision a future where healthcare services are not only more accessible and efficient but also deeply personalized and compassionate. Through our collective efforts, we strive to usher in an era where the boundaries between technology and healthcare blur, paving the way for a healthier and more connected society.

1.2 Motivation

The crucial need to transform healthcare via the application of cutting-edge technology and data-driven insights is what motivated us to work on this project. Recognizing the critical importance of early sickness detection and timely treatment, our goal has been to develop a predictive analytics platform that utilizes state-of-the-art machine learning techniques and is UI-integrated for user-friendliness.

Our platform is built on the Convolutional Neural Network (CNN) Sequential Model with SoftMax activation function, a potent algorithm renowned for its ability to extract intricate patterns and correlations from big datasets. With this cutting-edge technology, we want to analyze vast volumes of medical data, such as patient symptoms, medical records, and health indicators, to provide precise predictions regarding possible diseases.

Still, we are driven by more than just creating a forecasting tool. We firmly believe that everyone, regardless of technical skill or experience, should be able to utilize and benefit from our platform. Our priority to do this is to create and deploy a user interface (UI) that is simple to use and makes our machine-learning model easily accessible. Users may quickly input their symptoms and obtain individualized forecasts by using intuitive navigation and simplified interactions.

Our motivation is further fueled by the belief that technology has the power to democratize healthcare and improve outcomes for individuals worldwide. By empowering individuals with the knowledge and tools they need to monitor their health and seek timely medical intervention, we envision a future where everyone has equal access to quality healthcare. Together, we are driven by a shared commitment to leveraging technology for the greater good and making a meaningful impact on the lives of individuals and communities around the globe.

1.3 Objective

- Provide a complete platform that combines machine learning methods for the analysis of different datasets about health indicators, symptoms, and medical history. To ensure data quality and consistency, use data preparation procedures to clean and prepare the datasets for analysis.
- Create and train a CNN model that can analyze large datasets and identify important correlations and patterns in symptom data. Adjust the model's parameters to minimize the chance of overfitting or underfitting and maximize prediction accuracy.
- Create an intuitive user interface for the predictive analytics platform by utilizing Django, a high-level Python web framework. Use user experience (UX) techniques and simple design ideas to make sure users can navigate while interacting with the site easily.
- As the platform expands, set up scalable architecture and infrastructure to handle growing data volumes and user traffic. Use thorough testing procedures to find and fix any possible performance issues and guarantee platform dependability.
- Integrate channels of communication, such as chat functions, to let users and medical experts communicate directly. Permit users to request individualized medical guidance and support based on their symptoms and suspected illnesses.
- Put the predictive model through a rigorous testing program to evaluate its precision, dependability, and efficacy in predicting illness. To assess the model's effectiveness and dependability, compare its predictions to recognized medical diagnoses and results.
- Request input from users and consider feature requests to improve the platform's usability and usefulness over time. Update and enhance machine learning models and algorithms often to increase prediction precision and adjust to evolving healthcare trends.
- Work together with healthcare organizations and providers to guarantee that the platform conforms with legal requirements and regulations, including HIPAA

(Health Insurance Portability and Accountability Act). Use strong security measures, such as encryption and access controls, to protect patient privacy and confidentiality.

- Offer users educational materials and data about the significance of early illness diagnosis and proactive healthcare administration. Encourage consumers to actively participate in their health by raising their knowledge of available treatments and preventative strategies.

1.4 Problem Statement

The primary objective of our project is to create a symptom-based, machine-learning-powered illness prediction system. Our approach attempts to provide people with actionable insights and individualized suggestions based on their reported symptoms by utilizing the abundance of data on symptoms and related disorders that is already accessible. Users will be able to input their symptoms via a simple-to-use interface, and our machine-learning model will process and evaluate the data. Our system's core competency is its ability to leverage pattern recognition and predictive modeling. The system will go through enormous volumes of symptom data using complex algorithms, finding subtle connections and patterns that might not be immediately obvious to human observers. The system we develop will be able to forecast probable illnesses by utilizing this huge amount of data, enabling users to take charge of their health and seek the right medical attention when needed. In addition, our method aims to improve healthcare access by removing historical obstacles and giving people quick and simple access to medical guidance and support. Our technology is designed to function as a virtual healthcare gateway, providing users with personalized recommendations and insights based on their requirements and circumstances, regardless of their location—whether they are in busy urban areas or distant rural locations.

1.5 Approach

- **Data Collection and Preparation** : Assemble a varied dataset that includes different CSV files for both training and testing purpose and that CSV files contains data related to the different symptoms and its related disease. In order to guarantee consistency and improve model generalization, preprocess the CSV file using Label Encoding, Null Value Check, Convert data into tensor.
- **Model Building and Architecture** : Create a personalized Sequential Convolutional Neural Network (SCN) architecture leveraging SoftMax activation functions after three convolutional layers. To reduce computational complexity and down sample feature maps, add max-pooling layers after each convolutional layer. To minimize overfitting and boost model generalization, include dropout layers.
- **Training and Validation** : Separate the preprocessed dataset (for example, using a 70:15:15 ratio) into training, validation, and testing sets. Applying the training set and the proper operator (such as Adam, SGD) and loss function (such as binary cross-entropy, categorical cross-entropy), train the CNN model. Adjust hyper parameters and avoid overfitting by validating the model's performance with the validation set.
- **Model Evaluation and Optimization** : Leveraging the testing set, examine the trained model to determine its accuracy, precision. Analyze the model's performance in predict different diseases using methods like confusion matrices. To improve the model's capacity for prediction, optimize its architecture and hyper parameters in light of the evaluation's findings.
- **Forecasting and Implementation** : To enable multi-class detection, link the trained CNN model for different type of diseases detection for creating a prediction model. Provide a user-friendly online portal where users can input symptoms and get the knowledge about the probable diseases and get the consultation from the medical expert. Install the prediction model on a web-based platform so that it may be easily accessed and integrated with clinical procedures.
- **Feedback and Ongoing Improvement** : To boost the model's robustness and accuracy, keep it updated with fresh information and user and healthcare

professional input. Work together with experts in the field for incorporating domain-specific information and perspectives into the model for improved Symptoms based disease detection performance. Keep a close eye on the model's performance to make sure it works well in actual clinical settings.

1.6 Scope of the project

- Using CNN sequential models for symptom-based disease prediction, which makes it possible to categorize and identify diseases according to symptoms that have been recorded.
- Creating specific CNN models with high prediction accuracy designed to evaluate symptom data and forecast a range of diseases.
- Using CNN output-based prediction algorithms, users can receive individualized disease predictions based on symptoms they have reported.
- Using a simple-to-operate online interface to deploy the prediction model, enables users to input their symptoms and get real-time disease predictions.
- Including feedback mechanisms and user-friendly interface components to collect user data and iteratively improve the prediction model based on user input.
- Ensuring the platform's scalability and accessibility to support a diverse user base and offer disease prediction services to people everywhere.
- Carrying out thorough testing and validation processes to confirm the prediction model's reliability and precision in correctly detecting diseases based on reported symptoms.
- Working together with medical experts to include the symptom-based disease prediction system into clinical workflows to promote its uptake and incorporation into medical procedures.
- Upholding moral standards at every turn, including informed consent, patient confidentiality, and data protection, to guarantee moral research and project execution.

1.7 Organization of the Rest of the Report

The thesis project will be organized into several sections to provide a comprehensive overview of the Symptoms based disease prediction project consisting of five main sections: Literature Review, Methodology, Testing of the Model, Results and Discussion, and Conclusions and Future Scope.

- **Literature Review** : This section will critically review existing research pertinent to symptoms-based disease prediction using ML models. It will synthesize findings from relevant studies, identify gaps in current literature, and delineate the rationale for the comprehensive project. Seven selected papers will be examined to contextualize the project within the broader landscape of diseases detection methodologies.
- **Methodology** : The Methodology section will elucidate the research design, data acquisition, and analytical methods employed in the development of the symptoms-based diseases detection system. Detailed explanations will be provided regarding the utilization of CNN sequential models, data preprocessing techniques, and model training procedures. Additionally, the process of model evaluation and validation will be delineated to ensure the reliability and accuracy of the developed system.
- **Testing of the Model** : This section will outline the rigorous testing protocol undertaken to assess the performance of the CNN sequential models in detecting different type of diseases like Fungal infection, Chronic cholestasis, Hypertension, etc. Test scenarios, evaluation metrics, and encountered challenges will be meticulously documented. Furthermore, the results of testing, including performance metrics and any encountered issues, will be discussed in detail.
- **Results and Discussion** : Findings from the project will be presented in this section, alongside a comprehensive evaluation of the developed symptoms-based disease prediction system. Unique features and functionalities of the system, such as prediction capabilities, will be thoroughly examined. Additionally, implications of the results for particular disease and treatment will be discussed, providing insights into the potential impact of the project on healthcare outcomes.

- **Conclusions and Future Scope :** This section will summarize key findings from the project and offer recommendations for future research and development endeavors. The potential impact of the developed Symptoms based disease prediction system on improving early diagnosis and patient outcomes will be discussed. Furthermore, future directions for enhancing the system's capabilities and incorporating user feedback will be outlined.
- **References :** The References section will provide a comprehensive list of all sources cited throughout the report, adhering to the prescribed citation style guidelines. This section will ensure transparency and scholarly rigor in acknowledging the contributions of prior research to the project.

Chapter 2

Literature Review & Problem Identification

2.1 Literature Review

The purpose of the work was to use a weighted KNN algorithm to develop a disease prediction system. The system predicts over 230 diseases by taking several important variables including age, gender, and symptoms. It is shown how useful this prediction system could be for disease diagnosis, especially in situations where resources are limited, like the COVID-19 pandemic. The system that was developed achieves a 93.5% accuracy rate during testing and validation. More diseases and symptoms related to it were added to dataset to improve the scope of the system. Adding a wider range of symptoms to increase scope and prediction precision. [1]

In this research, various machine learning (ML) algorithms were used like Naïve Bayes, Decision Tree, and Random Forest for study. Based on a dataset of 4920 patient records with 41 diseases and 95 symptoms, the study tries to evaluate their effectiveness in disease prediction. The study emphasizes the value of AI in medical data analysis and shows how it can improve the precision of disease prediction and guide clinical decision-making. 95.12% is the mentioned accuracy achieved by the ML algorithms. [2]

With the aim predict diseases based on user-provided symptoms, this study uses a variety of machine learning (ML) methods, such as Decision Tree, Random Forest, KNN, and Naïve Bayes. The New York Presbyterian hospitals provided the dataset utilized in this study. A complete collection of features for analysis and prediction is provided by the testing dataset, which has 5 rows and 133 columns, whereas the training dataset has 4920 rows and 133 columns. In order to create a simple interface for recording user symptoms and assisting in disease prediction, the research made use of the Tkinter interface. The disease prediction system's remarkable 97.61% accuracy shows how effective machine learning algorithms are in correctly predicting diseases from user-provided symptoms. [3]

Using various kinds of machine learning methods, such as Random Forest, LSTM (Long Short-Term Memory), and SVM (Support Vector Machine), this research proposes a disease prediction model for humans. By combining practical problems data and medical datasets with weighted values based on rareness, the model outperforms previous approaches in terms of accuracy and reliability. The accuracy of the model is 97%. One significant void in the literature reviewed in this research is how little consideration is given to the possible obstacles and constraints involved in applying the suggested illness prediction model in actual healthcare environments. Important factors that need more research and discussion include practical issues like security of data, regulatory compliance, and seamless interaction with current healthcare systems. [4]

A number of machine learning (ML) techniques, such as CNN (Convolutional Neural Network), KNN (K-Nearest Neighbors), Naïve Bayes, Decision Tree, and Logistic Regression, are used in this study's proposed approach for predicting chronic diseases. The Nationwide Inpatient Sample (NIS) dataset was analyzed in this study. Enabling early diagnosis of chronic diseases will lower related risks and improve patient outcomes, which was the main objective of this research. The findings show that the suggested system achieves a 95% accuracy level. working together with organizations and healthcare professionals to apply the prediction models in clinical situations. The purpose of this collaboration is to improve the models' usefulness and validate their efficacy in actual-life situations. carrying out prospective research to evaluate how early disease identification affects patient outcomes and medical expenses. The results of these studies will be very helpful in understanding the financial and medical benefits of using predictive models in healthcare settings. [5]

This study demonstrates the need of quickly predicting diseases based on their symptoms in order to take the appropriate preventative measures to protect oneself from various illnesses. For the predictive analysis of the subject in this study, they implemented a variety of machine learning methods, including XGBoost, Random Forest classifier, SVM, and KNN classifier. However, KNN classifiers provided the highest accuracy when compared to other classification techniques. The corresponding accuracy values for

XGBoost, Random Forest, SVM, and KNN classifiers were 71%, 77%, 87%, and 97%. The research's future scope includes scanning pathology report photos so that an ML model may provide a diagnosis by taking into account various criteria. However, because it is difficult to obtain the scanned pathology reports from any data repositories, the construction of the dataset containing the pathology reports presents the issue in this instance. [6]

This research aims to identify trends in disease prediction across different types of supervised machine learning models through a performance measure evaluation. Supervised machine learning (ML) algorithms have demonstrated the ability to outperform standard disease diagnosis techniques and help physicians identify risk illnesses early. The Naïve Bayes classifier, Random Forest algorithm, and Support Vector machine are the algorithms utilized. We compare each algorithm's performance using K fold cross validation. It demonstrates that the Random Forest and Nave Bayes techniques are marginally more accurate than the other one. The above study achieves an accuracy of 96%. [7]

2.2 Problem Identification

One of the most common challenges in disease prediction studies is the widespread class imbalance problem in medical datasets. This mismatch can distort model projections and risk their credibility, with some disease classes being underrepresented in comparison to others. Innovative strategies including the creation of false data and sampling methods, are being investigated in an effort to lessen this difficulty. By increasing the diversity of datasets and redistributing class distributions, these approaches can potentially strengthen prediction models against the negative impacts of class imbalance.

Alongside the complexities of algorithms, a significant challenge is translating research-driven insights into real-world healthcare applications. Strict attention to numerous factors, such as data security, regulatory compliance, and smooth connection

with the current healthcare infrastructure, is necessary for real-world application. Through careful and strategic navigation of these challenges, the work aims to close the gap between the theoretical efficacy of models and practical clinical utility.

Furthermore, it becomes clear that the usefulness and accessibility of disease prediction systems are critical factors in promoting their general adoption and effectiveness. Encouraging user engagement and improving overall system efficacy require customizing user interfaces to meet the varied demands of patients and healthcare providers. The work aims to improve disease prediction systems accessibility and usability through iterative design and user-centered methodologies, ultimately enhancing their influence on patient outcomes and healthcare delivery.

Moreover, expanding the scope of studies to include a wider range of symptoms and diseases requires careful collection and compilation of extensive information. Obstacles in obtaining datasets, especially in specialist fields like pathology reports, highlight the necessity of teamwork and creative approaches to obtaining data. By overcoming these obstacles, the work aims to improve the disease prediction model's predictive power and increase the range of clinical scenarios in which they can be used.

Determining the effectiveness and appropriateness of machine learning algorithms for disease prediction tasks requires thorough evaluation. The work aims to identify the advantages and disadvantages of different algorithms by thorough testing and performance evaluation, enabling academics and practitioners to choose and implement algorithms with knowledge.

Finally, for prediction models to be validated and used in actual clinical settings, collaborations with medical organizations are necessary. The work aims to place predictive models within the complex environment of medical practice by developing healthy partnerships with stakeholders across the healthcare ecosystem and clarifying their concrete influence on patient outcomes and healthcare expenses.

Chapter 3

Methodology

3.1 Comparison of Machine Learning Models

- **Decision Tree** : Decision trees are a type of flowchart-like structure in which each internal node represents a "test" on an attribute, each branch represents the test's outcome, and each leaf node represents a class label. They are helpful for understanding the reasoning behind classification decisions because they are simple to understand and intuitive, but they can be prone to overfitting, particularly when dealing with complex datasets.[8]
- **Support Vector Machine (SVM)** : SVM is an effective supervised learning technique that may be applied to regression, outlier identification, and classification. SVM works by locating the perfect hyperplane in a high-dimensional space that optimally divides the data points of various classes. Since it defines the decision boundary using only a fraction of training points (support vectors), it is memory economical and effective in high-dimensional spaces. SVM may use strategies like the kernel trick to address non-linear decision boundaries.[9]
- **Random Forest** : Random Forest is an ensemble learning technique that builds a more accurate and reliable model by combining many decision trees. Random Forest introduces randomness in the selection of data samples (bootstrap samples) and the features considered at each split, leading to diverse trees that collectively provide better generalization. Unlike a single decision tree, which can overfit the training data, Random Forest reduces overfitting by averaging the predictions of many trees trained on random subsets of the data. Large datasets and high-dimensional data are two things that Random Forest excels at handling, and it can sustain strong performance without a lot of hyperparameter tweaking.[10]
- **Gradient Boosting** : A machine learning ensemble approach called gradient boosting successively combines the predictions of several weak learners, usually decision trees. By refining the model's weights based on the mistakes of earlier

iterations, it seeks to enhance overall predictive performance by progressively lowering prediction errors and raising the model's accuracy.

- **Sequential Model** : Neural networks frequently utilize a sequential model with dense (completely connected) layers and linear layers for a variety of tasks, such as regression, classification, and other machine-learning tasks. The model is organized in this design as a linear stack of layers, with connections between each layer and just the ones that come before and after it. In this design, dense layers also referred to as completely linked layers are essential building blocks. The model can identify intricate patterns in the data because every neuron in a dense layer receives information from every other neuron in the layer above it. Implemented in PyTorch and other deep learning frameworks, linear layers multiply the input data by a weight matrix and add a bias term to accomplish a linear transformation. Non-linear activation functions, such as ReLU (Rectified Linear Unit), usually follow these layers. They provide non-linearity to the model and help it discover intricate links in the data. Once considered, a sequential model with dense and linear layers is a flexible architecture that can be trained to optimize its parameters for a particular learning task using methods like gradient descent and backpropagation.

3.2 Proposed Methodology

- **Sequential Model Structure** : Personalized Sequential Convolutional Neural Network (CNN)
- **Overview of Sequential Model** : For efficient feature extraction and symptoms-based disease prediction applications, the proposed CNN architecture makes use of sequential layers of linear layers. This method is easy to understand and often applied, particularly for basic feedforward neural networks in which information moves from the input layer to the output layer via hidden layers.
- **Linear Layer** : A linear layer, also known as a fully connected layer, is one of the fundamental building blocks in neural networks. It consists of a set of neurons where each neuron is connected to every neuron in the previous layer. In this case, each input to this layer has 132 features after passing through these layers, the

output will have 41 features. In order to extract complex structures and representations from the input data and utilize them to solve tasks or make predictions, a linear transformation is necessary.

$$\text{Output} = \text{Input} \times \text{Weights} + \text{Bias}$$

- **SoftMax Activation Function :** As the last layer in a neural network architecture, the SoftMax activation function is used to transform the raw output scores generated by the layers before it into a probability distribution over the classes. This makes it easier to understand the model's predictions because each output corresponds to a class's probability. Additionally, because the model's output can be directly compared with the true class labels during training, SoftMax makes it easier to calculate loss and optimize.

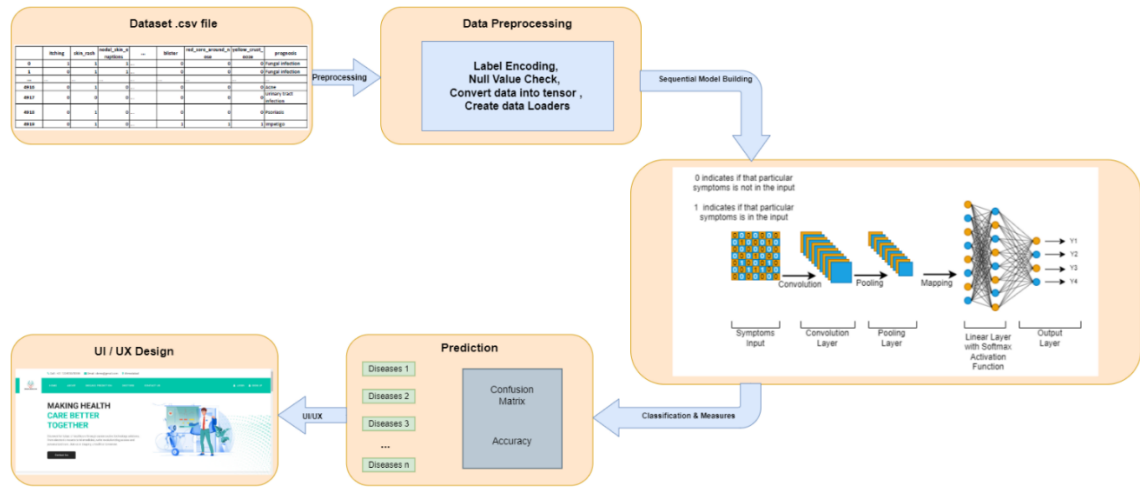


Figure 3.1: Architecture Of Proposed Methodology

3.2.1 Data Collection and Preprocessing : Search the data set of the symptoms that can be used for the disease prediction model from the Kaggle, GitHub, Google Dataset Search, etc. To improve dataset variability, preprocess the CSV files by Label Encoding, Null Value Check, and Convert data into a tensor.

3.2.2 Model Architecture Design : To extract spatial characteristics from the input data, design a Convolutional Neural Network (CNN) architecture

that consists of convolutional layers followed by linear layers. Run three sets of convolutional layers with different filter widths and the input features are 132 and output features are 41. The output of the final convolutional layer is flattened and then sent through fully linked dense layers, and SoftMax activation functions to add non-linearity. To reduce overfitting during training, use dropout layers, which randomly deactivate neurons after each thick layer.

3.2.3 Training and Validation : Using a ratio of 80:20, divide the preprocessed dataset into training and testing sets. Using the training set, train the CNN model with the proper learning rate, optimizer (such as Adam, SGD), and loss function (categorical cross-entropy for multi-class classification, or binary cross-entropy for binary classification). Utilizing the validation set, assess the model's performance by tracking metrics like accuracy, and loss precision. Then, modify the hyper-parameters to achieve the best performance.

3.2.4 Model Evaluation : To determine the resilience as well as the adaptability of the model that was taught across the testing set, evaluate its performance. Calculate assessment measures including recall, accuracy, and precision to determine how well the model performs in symptoms-based disease prediction using various modalities.

3.2.5 Prediction Model Development : To enable multi-class classification, integrate the CNN model that has been trained for different disease classifications (Fungal Infection, Allergy, Chronic Cholestasis, etc.) to create a prediction model. Provide a user-friendly interface for the prediction model that enables users to add their symptoms and from their input, it helps to predict the disease.

3.2.6 Deployment and Integration : Put the prediction model on an online platform that users may access and use with convenience. Integrate the prediction model with telemedicine applications or modern healthcare systems to incorporate it into clinical processes efficiently. Enforce data security and privacy rules strictly to ensure compliance with relevant laws and protect patient privacy.

3.2.7 Continuous Improvement : For bettering the prediction model's reliability and precision, add new data to it frequently and make architectural adjustments in response to performance reviews. Work together with healthcare professionals and domain specialists to integrate domain-specific knowledge and insights into the model, hence enhancing its effectiveness in predicting different diseases based on its symptom.

Chapter 4

Result And Discussion

4.1 Model Training, Testing, and Prediction

To evaluate the effectiveness of the developed CNN model, the confusion matrix plot has been used. Furthermore, the subsequent metrics were accurately computed: Accuracy can be measured as the ratio of correct predictions to total predictions.

$$\text{Accuracy} = \frac{(TP+TN)}{(TP+FP+FN+TN)}$$

The ratio of true positives to all expected positives is known as precision.

$$\text{Precision} = \frac{TP}{(TP+FP)}$$

True positive, false positive, false negative and true negative values are the output metrics for the training and validation symptoms of the models, denoted by the letters TP, FP, FN, and TN.

4.2 Result Analysis

Sequential Model:

These data completed 430 epochs of training (Early stopping occurs – Total 500 epochs) and a batch size of 32. In the last epoch, the model's training accuracy was 97.54%, and its validation accuracy was 97.66%.



Figure 4.1: Model Accuracy Versus Epochs Plot for Training and Validation

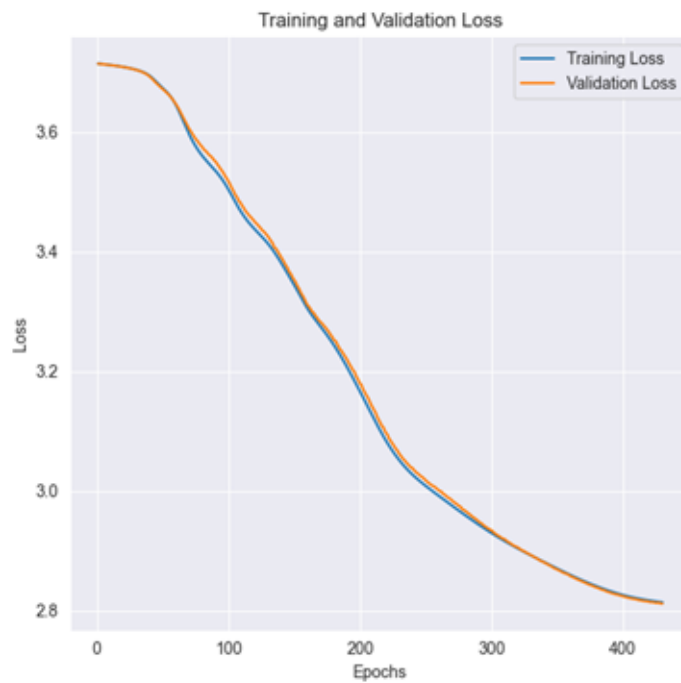


Figure 4.2: Model Loss Versus Epoch Plot for Validation and Training

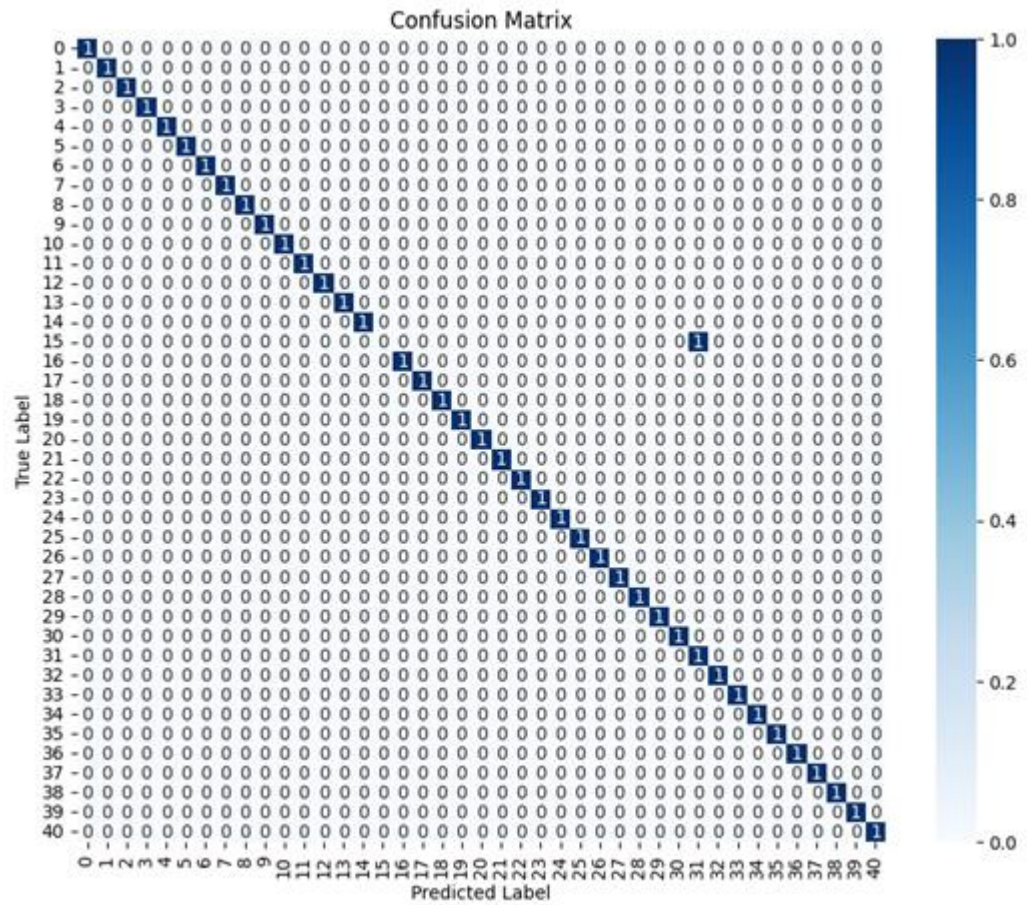


Figure 4.3: Confusing Matrix

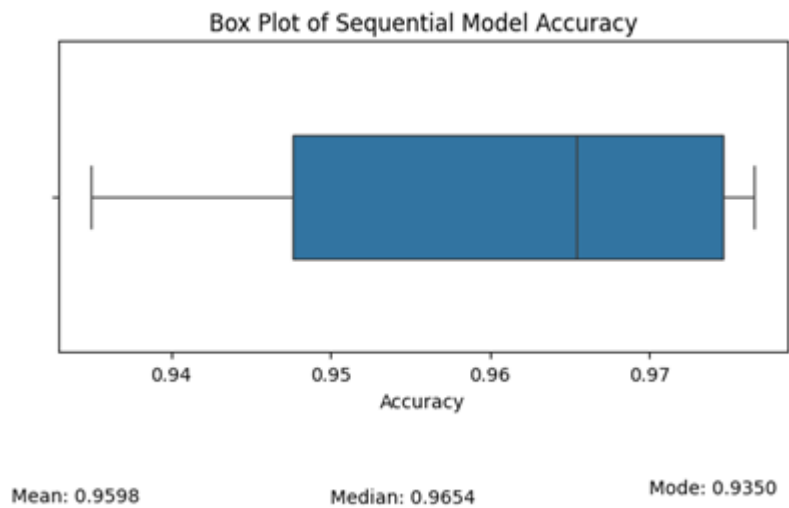


Figure 4.4: Box Plot for Sequential Model with SoftMax Activation Function

4.3 Statistical Analysis of Different Models

We evaluated the performance of various machine learning models including the Sequential Model with SoftMax activation function, Sequential Model with Dense Layers and Dropout Layers, Decision Tree, Random Forest, Support Vector Machine, and Gradient Boosting. Each model was iterated five times to ensure reliability, and the resulting metrics were plotted as box plots for comparison.

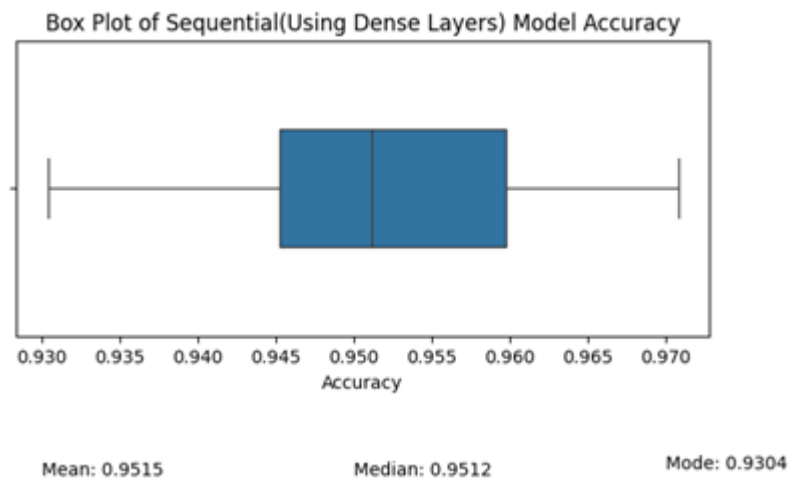


Figure 4.5: Box Plot for Sequential Model Using Dense and Dropout Layers

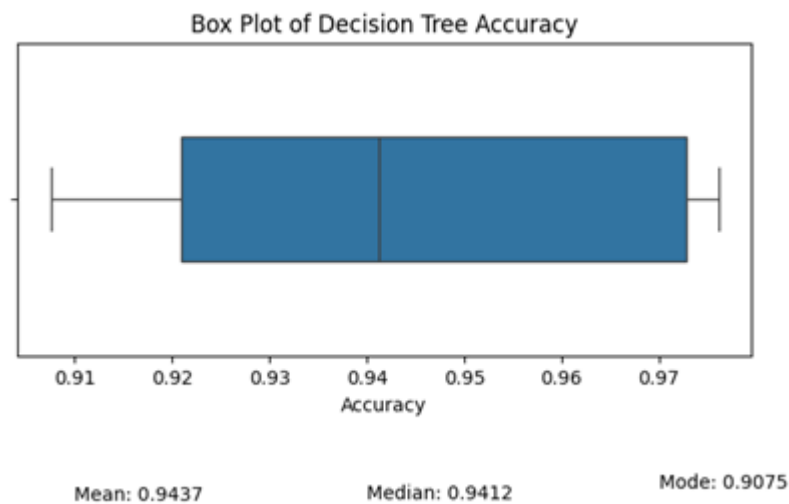


Figure 4.6: Box Plot for Decision Tree

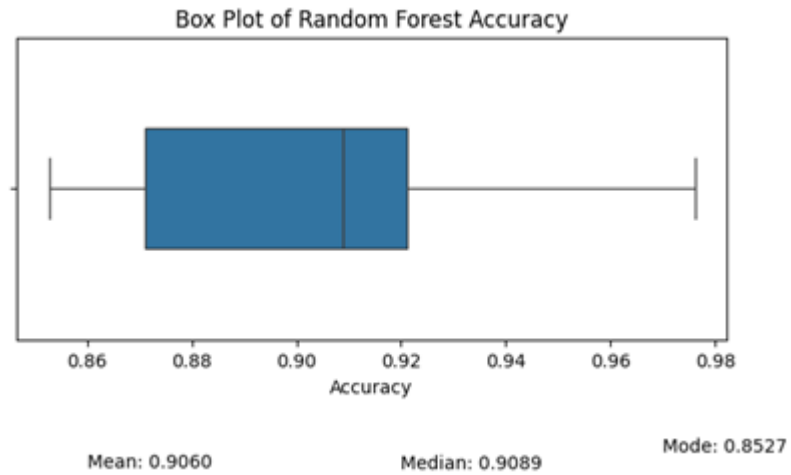


Figure 4.7: Box Plot for Random Forest

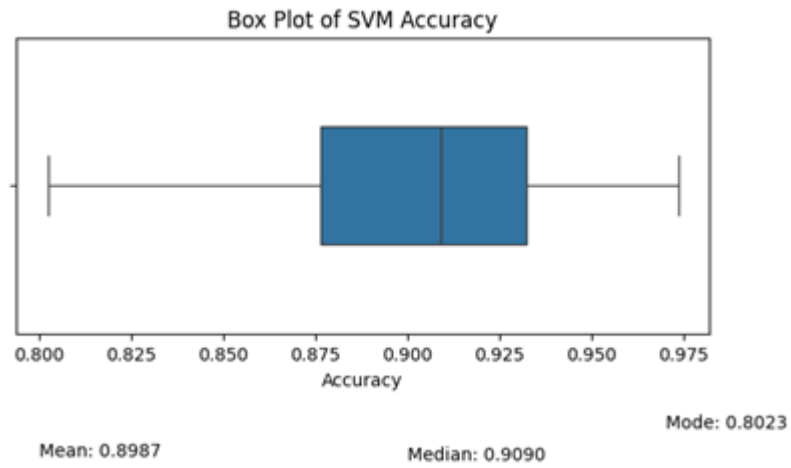


Figure 4.8: Box Plot for SVM

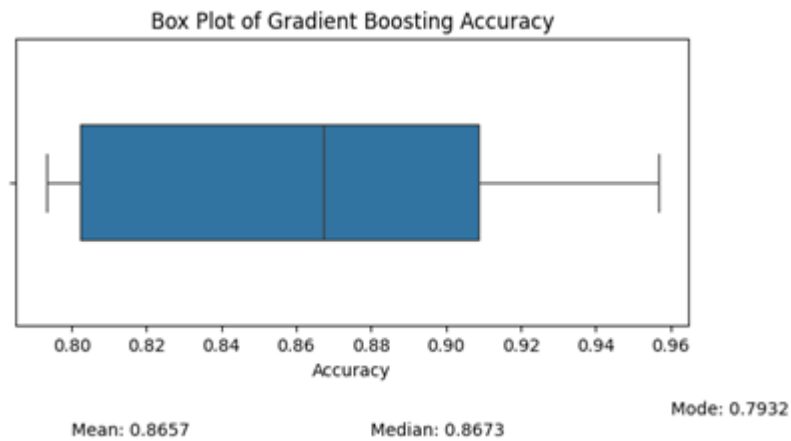


Figure 4.9: Box Plot for Gradient Boosting

4.5 Web Development Results

Here is a Flow chart for a better understanding of how the website works.

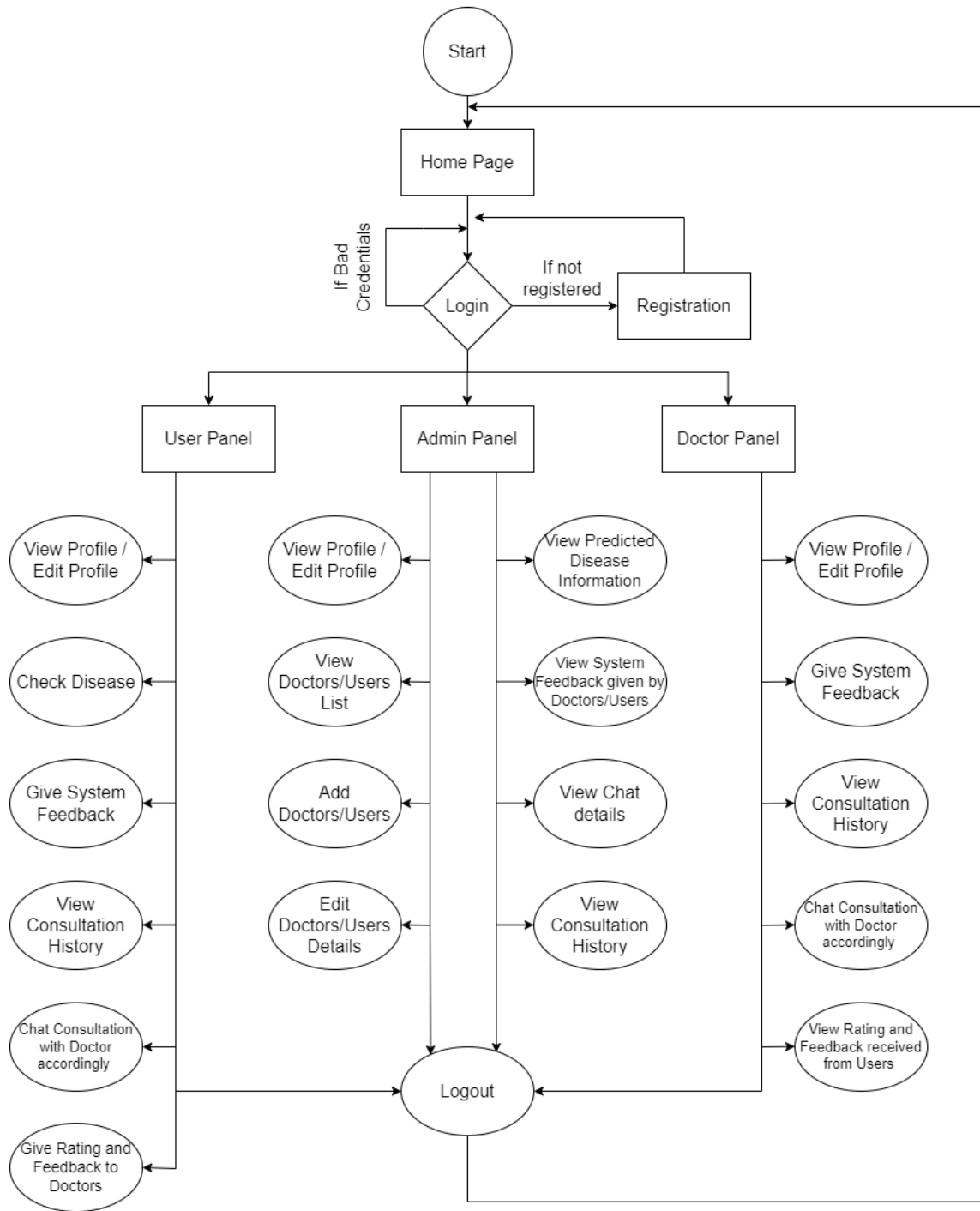


Figure 4.12: Flowchart of Website

Our web technologies output demonstrates the smooth integration of DL models with user interaction. When users enter symptoms, the website interacts with our deep-learning model to generate predictions. Users are empowered with important health insights due to the data's dynamic representation.

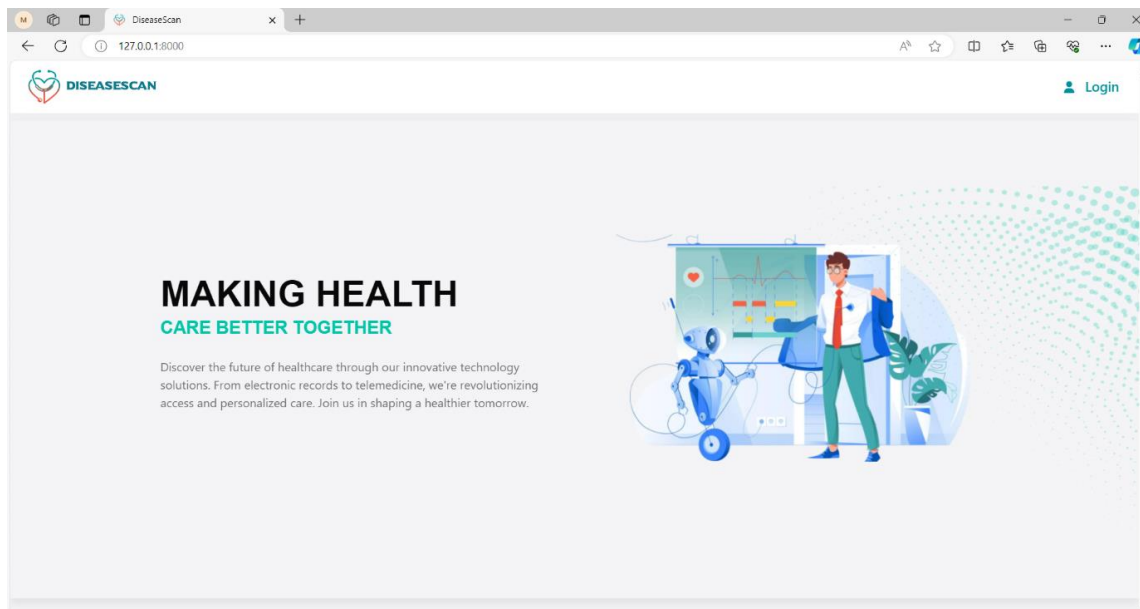


Figure 4.13: Home Page

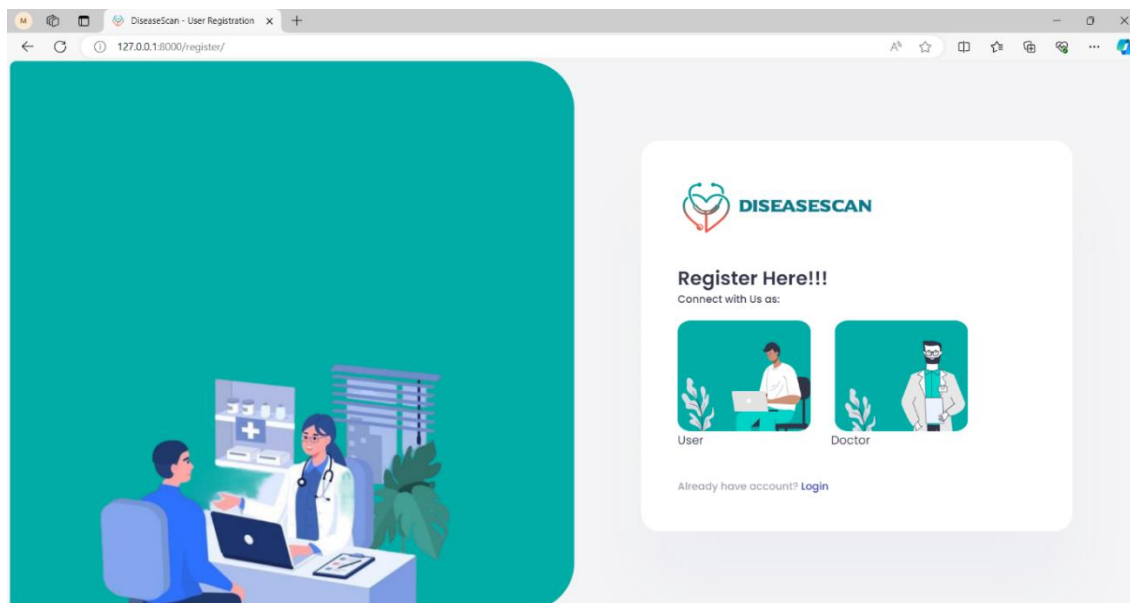


Figure 4.14: Registration Page(Select either User or Doctor)

DISEASESCAN

Register Here!!!

Start Journey for Better Health

User Name *
simoli20

Full Name *
Simoli Shah

Email *
simoli20@gmail.com

Date of Birth *
20-05-2006

Gender *
☐ Male ☒ Female ☐ Other

Mobile Number *
8200020069

User Image *
Choose File user-02.jpg

Password *

Figure 4.15: User Registration Page

DISEASESCAN

Login

Username *
simoli20

Password *

☐ Remember me [Forgot Password?](#)

[Login](#)

Need an account? [Sign Up](#)

Figure 4.16: Login Page(User/Doctor/Admin)

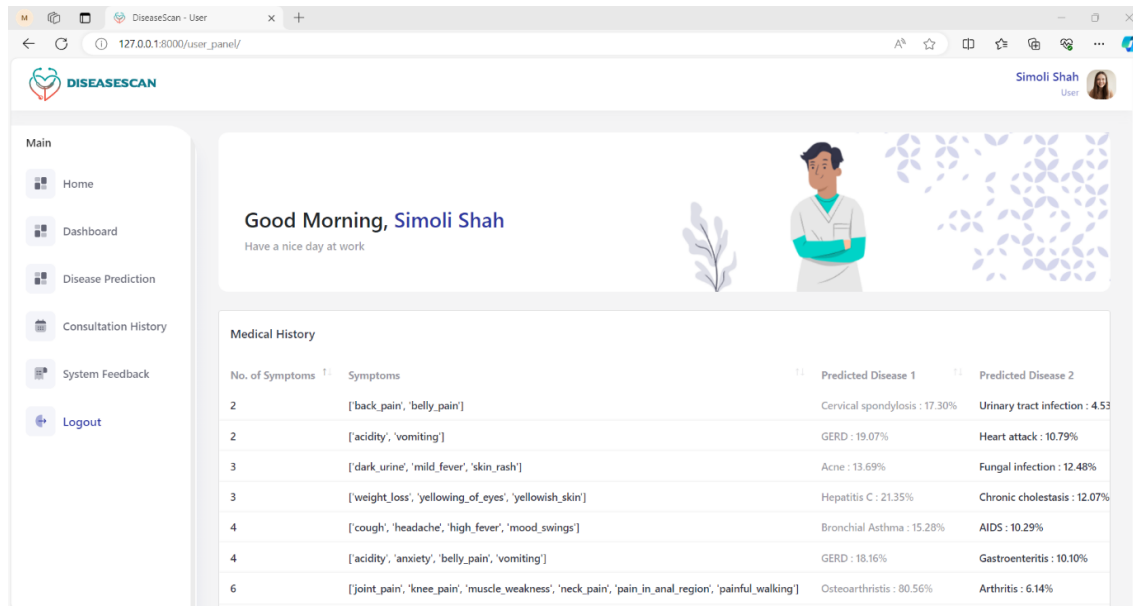


Figure 4.17: User Dashboard

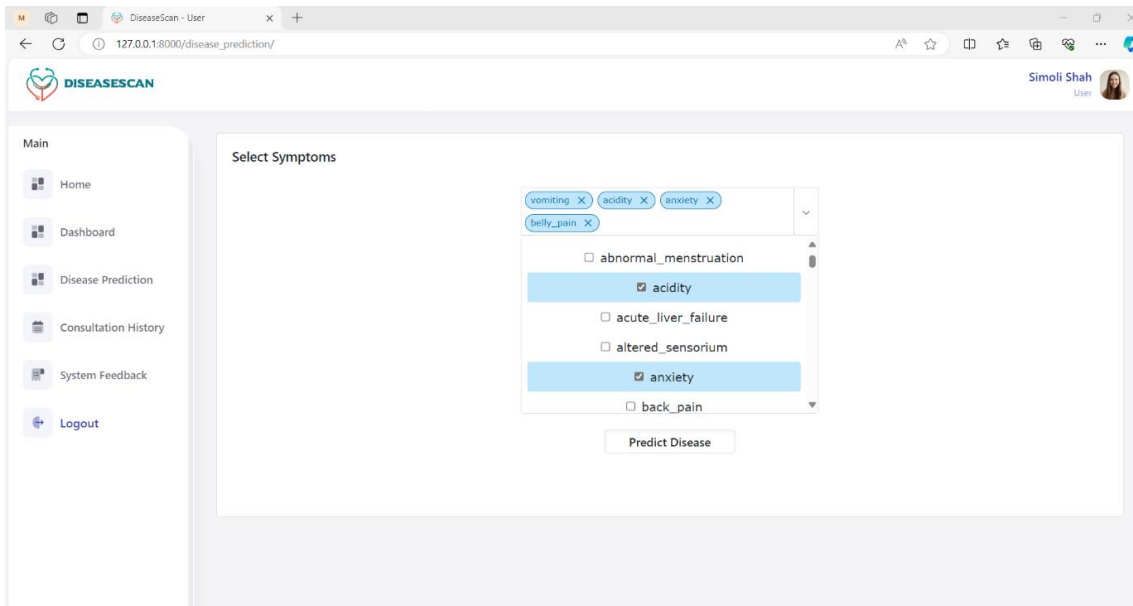


Figure 4.18: Symptoms Selection Page

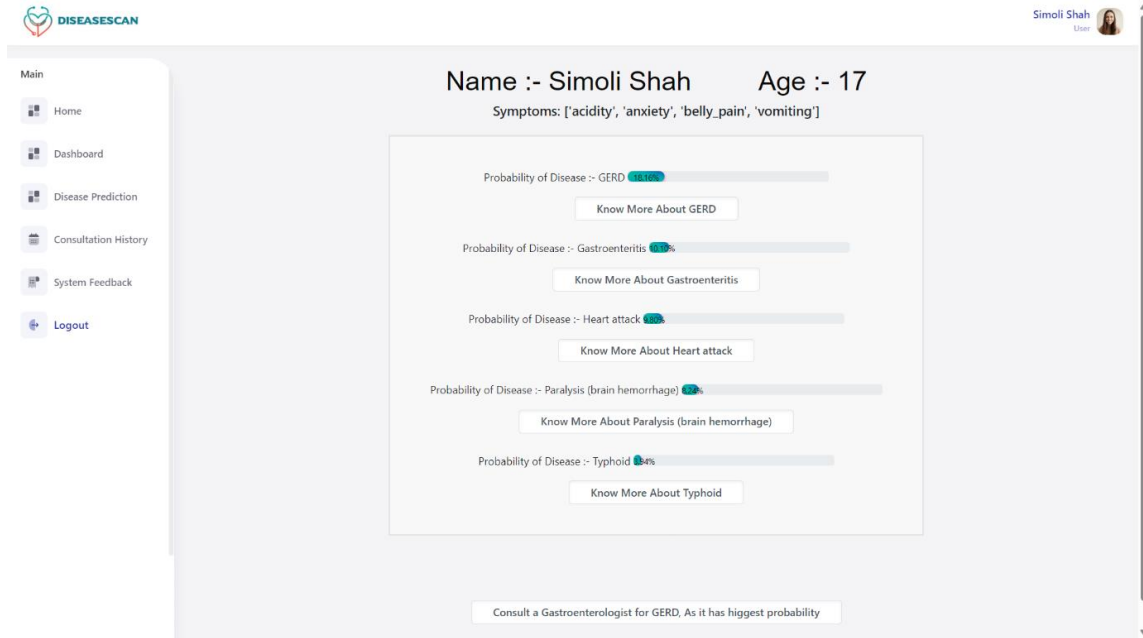


Figure 4.19: Predicted Disease Page

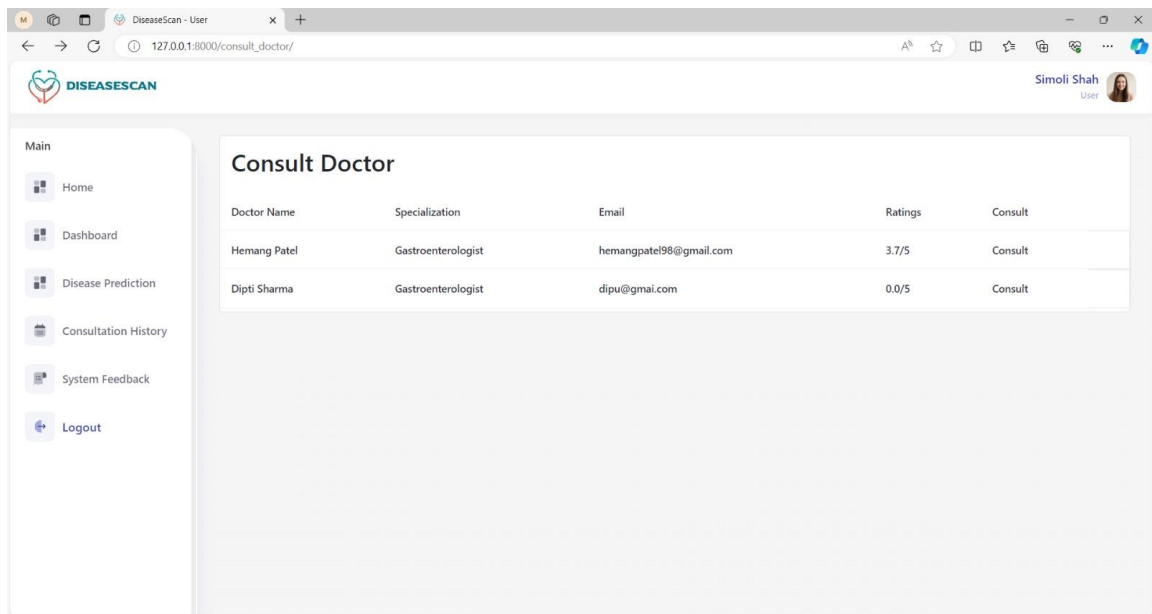


Figure 4.20: Consult Doctor Page

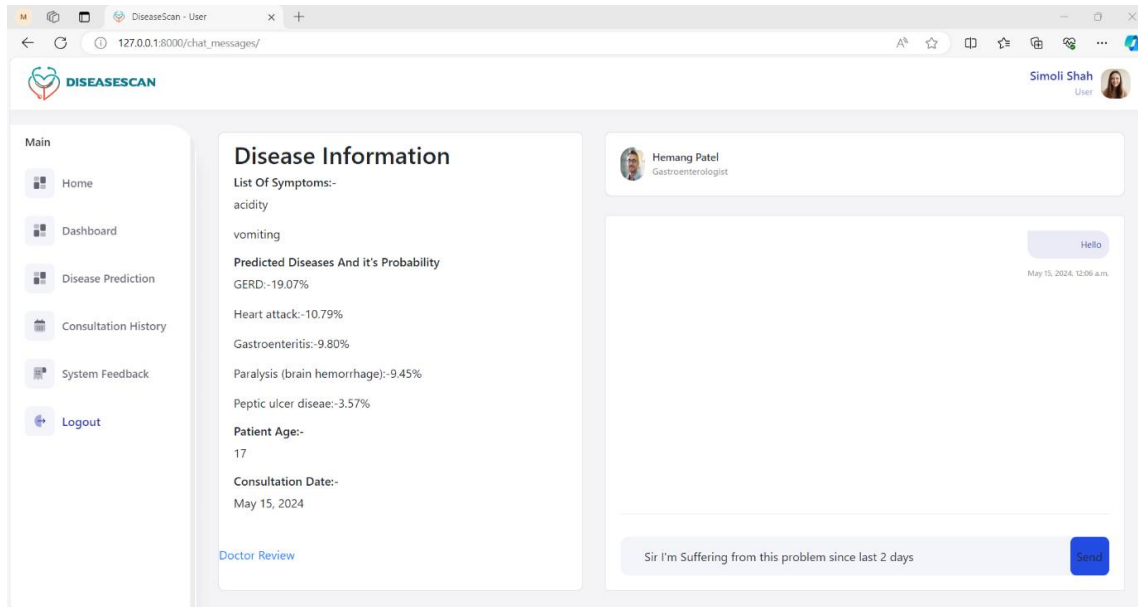


Figure 4.21: Chat with Doctor Page

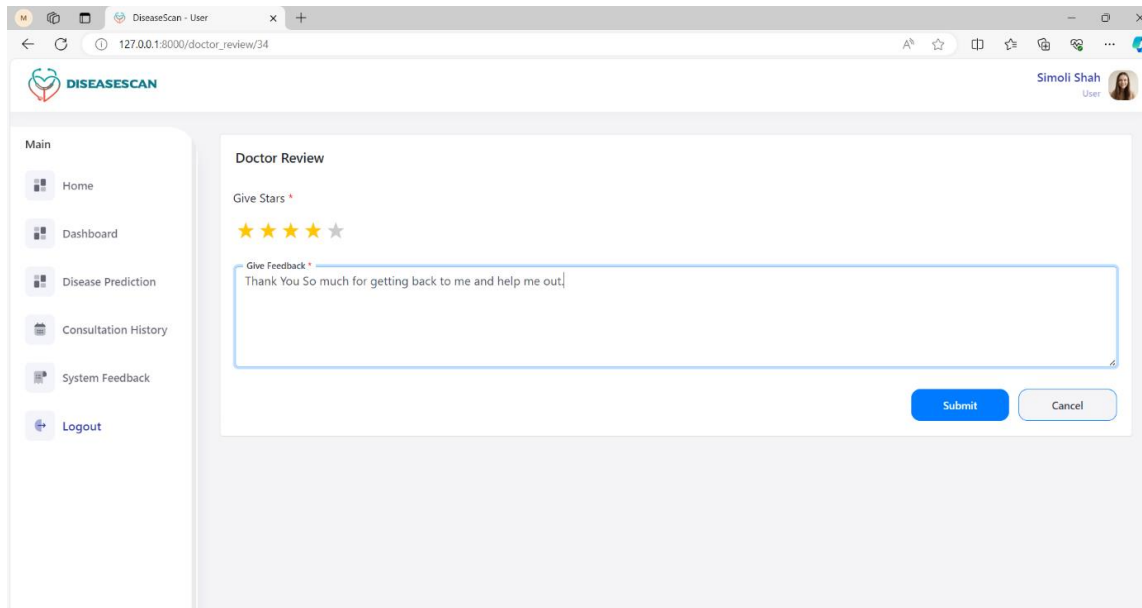


Figure 4.22: Doctor Review Page





Doctor Registration

Be Our Health Partner

User Name *
Hemang98

Full Name *
Hemang Patel

Email *
hemangpatel98@gmail.com

Date of Birth *
25-09-1975

Gender *
☒ Male
 ☐ Female
 ☐ Other

Mobile Number *
9223266882

User Image*
Choose File user-03.jpg

Registration Number *
453267

Qualification *
M.D

Specialization*
Gastroenterologist

Password *

Figure 4.23: Doctor Registration Page


Hemang Patel
Doctor

Dashboard

Consultation History

System Feedback

Logout

Good Afternoon, Hemang Patel

Have a nice day at work



Users Review

Patient Name	Rating	Review
Simoli Shah	4/5	Thank You So much for getting back to me and help me out.
Helly Vora	4/5	Nice Helping Nature.
Dharmesh Baldaniya	3/5	Good

Figure 4.24: Doctor Dashboard

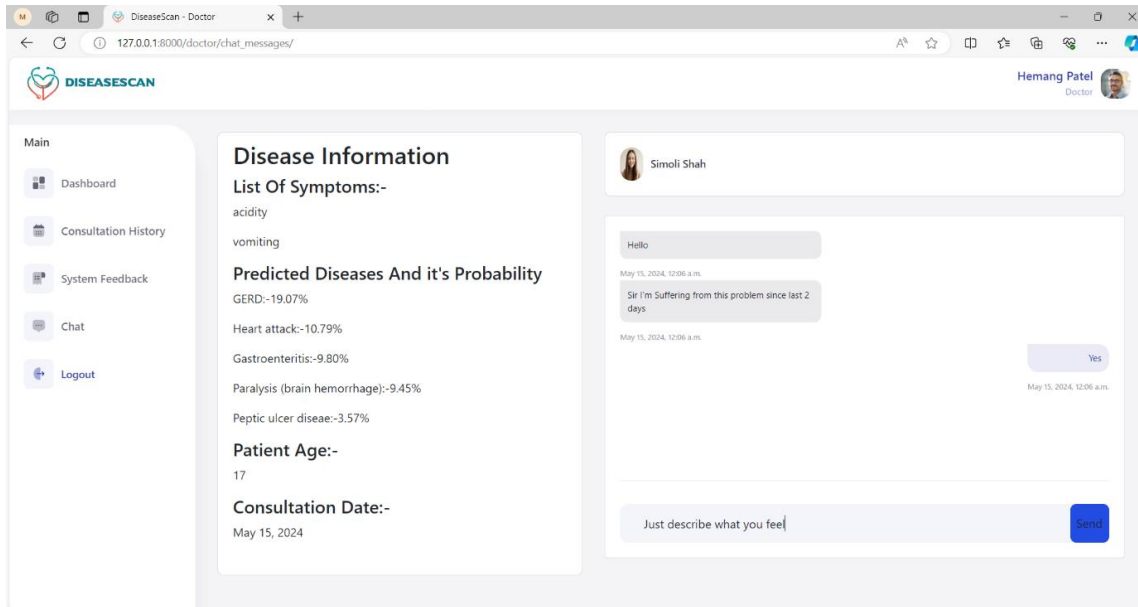


Figure 4.25: Chat with User Page

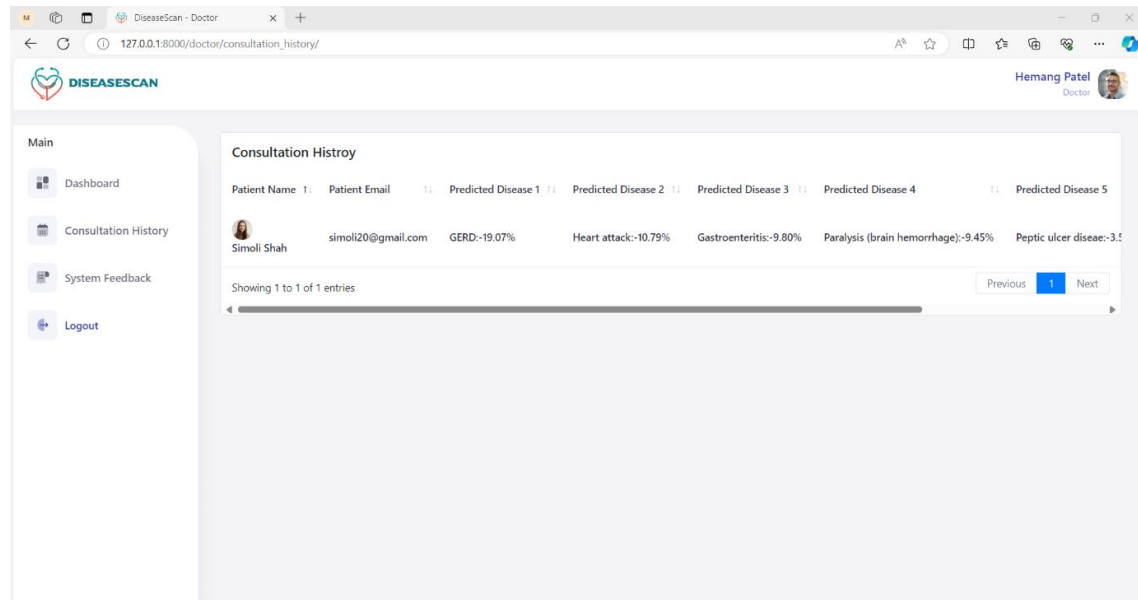


Figure 4.26: Consultation History Page(Doctor Panel)

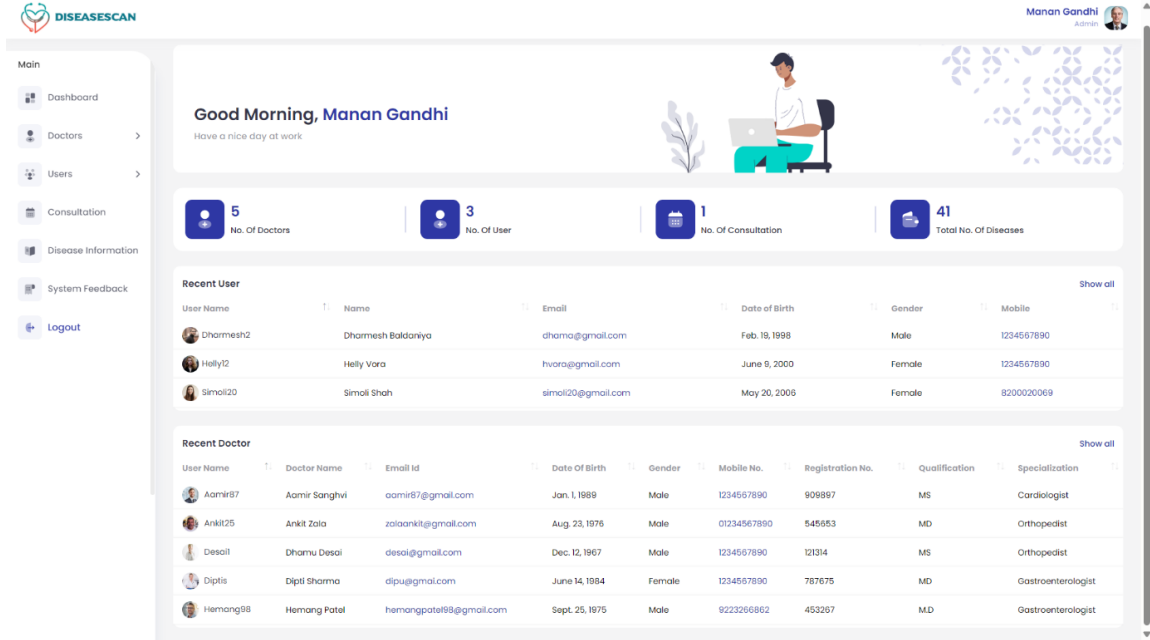


Figure 4.27: Admin Dashboard

4.6 Discussion

We explore the complexities and possible implications of our proposed symptom-based, machine-learning-driven disease prediction system. Fundamentally, our system stands for a proactive and individualized approach to disease management, marking an important change in the delivery of healthcare. Our system's capacity to access huge amounts of data on symptoms and associated diseases is one of its main advantages. Our system can sort through this data by using advanced machine learning techniques to find subtle patterns and correlations that could escape human observers. With earlier diagnosis and intervention possible, both of which are essential elements in enhancing patient outcomes, this ability has the potential to completely transform disease prediction. Furthermore, our system's focus on user interface design is crucial. Our goal is to revolutionize healthcare access by offering a user-friendly interface that makes it easy for people to enter their symptoms. Personalized healthcare advice and recommendations are available to people of different economic statuses and geographic locations. Users may feel more empowered to take charge of their health and seek prompt medical assistance when

needed as a consequence of this. There are many directions that further research and growth can go in the future. It will be crucial to fine-tune our machine-learning algorithms to improve usability and predictive accuracy. To promote wide adoption, further efforts should be taken to guarantee a smooth interface with the current regulatory frameworks and healthcare infrastructure. Furthermore, to guarantee that our system keeps up with the changing requirements of both users and healthcare providers, continuous assessment and improvement will be necessary. In summary, our suggested symptom-based, machine learning-driven disease prediction system has enormous potential to revolutionize the way healthcare is provided. Our goal is to increase access to healthcare, improve health outcomes, and ultimately allow people to live healthier lives by utilizing data and technology.

Chapter 5

Conclusion

After an in-depth examination of machine learning algorithms, including Support Vector Machine (SVM), Gradient Boosting, Random Forest, Decision Tree, and Sequential Models with Dense Layers and SoftMax activation function, for symptom-based disease prediction, the experimental findings clearly demonstrate the exceptional performance of the Sequential Model with SoftMax activation. Models like SVM, Gradient Boosting, Random Forest, and Decision Trees perform well in a range of situations, they have drawbacks including being difficult to comprehend, sensitive to parameter adjustment, and prone to overfitting. Additionally, Sequential Models with Dense Layers are good at capturing intricate patterns, but they are difficult to comprehend, overfit, and compute-intensive. The Sequential Model with SoftMax activation function, however, stands out as the most promising method despite these difficulties. This model uses Convolutional Neural Network (CNN) architecture to capture complex spatial correlations in symptom data, and it performs very well in both the training and validation stages. This model proved to be better in terms of both prediction accuracy and generalization ability throughout the experimental process by continuously outperforming its competitors in both the training and validation stages. Moreover, the SoftMax activation function added important non-linearity, which allowed the model to recognize intricate connections and patterns in the data. Combining CNN with SoftMax improved the model's interpretability and gave it the capacity to make consistent and dependable predictions in a range of clinical situations. As a result, it is safe to say that, after a thorough evaluation, the CNN Sequential Model with SoftMax activation function is the best option for symptom-based disease prediction operations. Its outstanding predictive performance and clinical usefulness will bring about a new era of precision medicine.

Chapter 6

Future Scope

- **Data Collection and Feedback Mechanisms** : Establish mechanisms for collecting user-provided data and feedback. Enable users to report the accuracy of predictions and provide suggestions for improvement. Utilize user feedback to refine the model and enhance prediction accuracy over time.
- **Model Adaptation and Continuous Learning** : Implement online learning techniques to adapt the model based on incoming data streams. Explore methods for updating the model parameters dynamically as new data becomes available. Incorporate mechanisms for model retraining or fine-tuning based on user feedback and evolving medical knowledge.
- **User-Centric Design and Transparency** : Prioritize user-centric design principles to ensure ease of use and accessibility. Provide transparent explanations of model predictions and decision-making processes to foster user trust. Incorporate features that allow users to understand how their data is being used and contribute to model improvement.
- **Enhanced Healthcare Decision-Making** : Aim to empower users to make informed healthcare decisions based on accurate predictions and personalized recommendations. Explore ways to integrate the model predictions into clinical workflows and decision support systems. Collaborate with healthcare professionals to validate the model's effectiveness in real-world settings and ensure its alignment with clinical standards.
- **Scalability and Performance Optimization** : Focus on optimizing the scalability and performance of the system to handle increasing user interactions and data volumes. Implement efficient algorithms and infrastructure to support real-time data processing and model adaptation. Explore cloud-based solutions and distributed computing architectures to enhance scalability and reliability.

- **Ethical and Regulatory Considerations :** Address ethical considerations related to user data privacy, consent, and security. Ensure compliance with relevant regulations and standards, such as GDPR (General Data Protection Regulation) and HIPAA (Health Insurance Portability and Accountability Act). Establish protocols for transparent communication with users regarding data usage, privacy policies, and security measures.

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